

Interview Project for ETL

Date: Week of 16 March 2026

Goal

A Transformer Service for Balance Sheet data

Technology

Language: Python (recommend using pandas and FastAPI) DB: use sqlite Repo: Place your code in github & share it prior to interview. We'll run the code and review your work

Transform

Implement a service that takes an excel file and converts it to a standard format output.

The standard format is three columns: **Account (string)**, **Last Day of Month (MM/DD/YYYY)**, **Amount (Number)**.

The input files (the 2 examples provided) have some header rows (like rows 1, 2 etc.), a row of column headers (example Periods). This is followed by rows of accounts and their amounts by period.

The 2 files differ in terms of tabs and other formatting information.

1. CIQ_Interview_Sheet_v1.xlsx - All accounts data starts at row 6. Accounts are in Columns A-D. An account has an Integer Prefix (e.g. 1020 BofA Checking 6759).
2. CIQ_Interview_Sheet_v2.xlsx - All accounts data starts at row 6. Accounts are in Columns A-D. An account has an Integer Prefix (e.g. 120 - Accounts Receivable).

This transform function should not aggregate. Skip all non-accounts.

The transform function needs to be able to handle these two formats and produce an output in the standard format in JSON.

Errors

If the input files are not in the expected format, then return an error.

1. Basic: A high level error that states the transform failed.
2. Intermediate: The detail on the 1st error it encountered.
 - a. Periods are not in the expected format
 - b. The rows are not in one of the 2 expected formats

3. Detail: All the errors encountered by row number & what went wrong

ETL Service

Implement a basic service using FastAPI that has the following endpoints

- POST API to create a new transformed standard formatted data
 - This takes the parameters
 - Name: is the name of the input file (example: BS-16Mar.xlsx)
 - Transformer: Can be either Transform1 (for the 1st example file) or Transform2 (for the 2nd example file).
 - Input file (xlsx) - its format is as described in the section above. The example files are two valid options.
 - This service will invoke the transform function (as described in the section above) and store the transformed output.
 - If the transform fails then return the error message otherwise a 200 with the id of the transformed file.
- GET API returns the list of all files that have been transformed
 - List in Json of the file name, file id & their create date
- GET API with file id. This returns a JSON of the file in standard format.
 - Note: Attached are expected output files
 - CIQ_Interview_Sheet_v1_Output.csv
 - CIQ_Interview_Sheet_v1_Output.csv
- DELETE API with file id. This deletes the transformed data file.